



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Sunday, September 20th, 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
- Summary of all results

Introduction

- Project background and context
- Problems you want to find answers

Section 1

Methodology

Methodology

Executive Summary

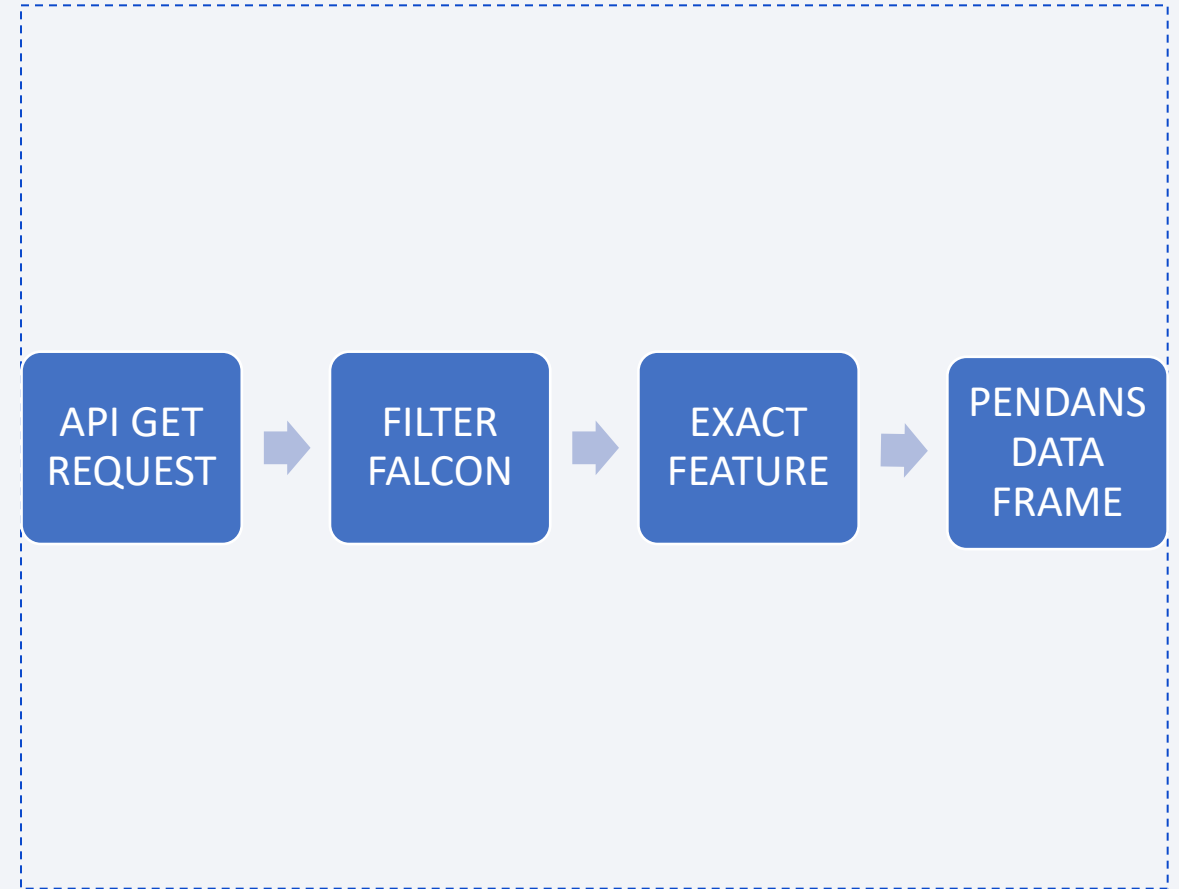
- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

- Describe how data sets were collected.
- You need to present your data collection process use key phrases and flowcharts

Data Collection – SpaceX API

- https://github.com/HE11032006/project_capstone_data_science_IBM/blob/main/1_Data_Collection_API.ipynb



Data Collection - Scraping

- https://github.com/HE11032006/project_capstone_data_science_IBM/blob/main/2_Data_Collection/WebScraping.ipynb

1 .Getting Response from HTML

```
page = requests.get(static_url)
```

2. Creating BeautifulSoup Object

```
soup = BeautifulSoup(page.text, 'html.parser')
```

3. Finding tables

```
html_tables = soup.find_all('table')
```

4. Getting column names

```
column_names = []
temp = soup.find_all('th')
for x in range(len(temp)):
    try:
        name = extract_column_from_header(temp[x])
        if (name is not None and len(name) > 0):
            column_names.append(name)
    except:
        pass
```

5. Creation of dictionary

```
launch_dict= dict.fromkeys(column_names)

# Remove an irrelevant column
del launch_dict['Date and time ( )']

launch_dict['Flight No.'] = []
launch_dict['Launch site'] = []
launch_dict['Payload'] = []
launch_dict['Payload mass'] = []
launch_dict['Orbit'] = []
launch_dict['Customer'] = []
launch_dict['Launch outcome'] = []
launch_dict['Version Booster'] = []
launch_dict['Booster landing'] = []
launch_dict['Date'] = []
launch_dict['Time'] = []
```

6. Appending data to keys (refer) to notebook block 12

```
In [12]: extracted_row = 0
#Extract each table
for table_number,table in enumerate(
    # get table row
    for rows in table.find_all('tr'):
        #check to see if first table
```

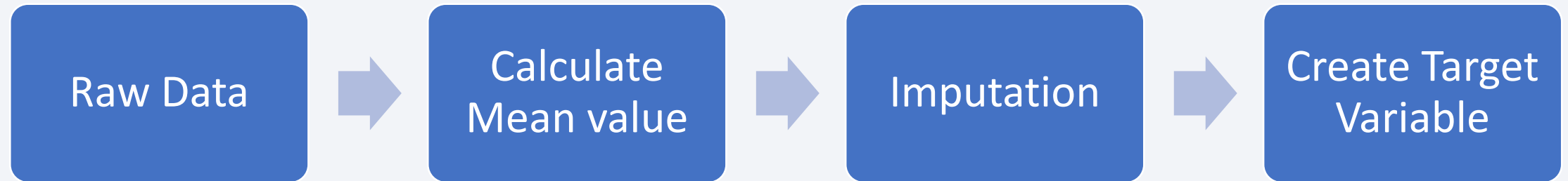
7. Converting dictionary to dataframe

```
df = pd.DataFrame.from_dict(launch_dict)
```

8. Dataframe to .CSV

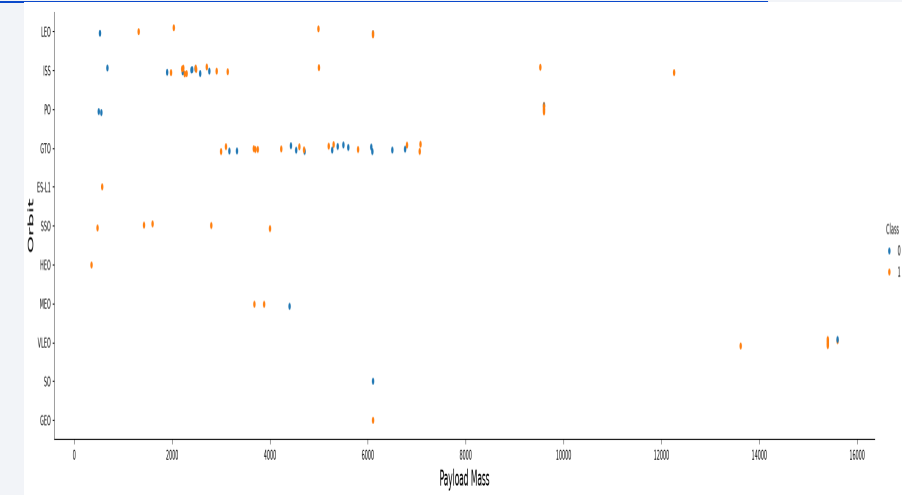
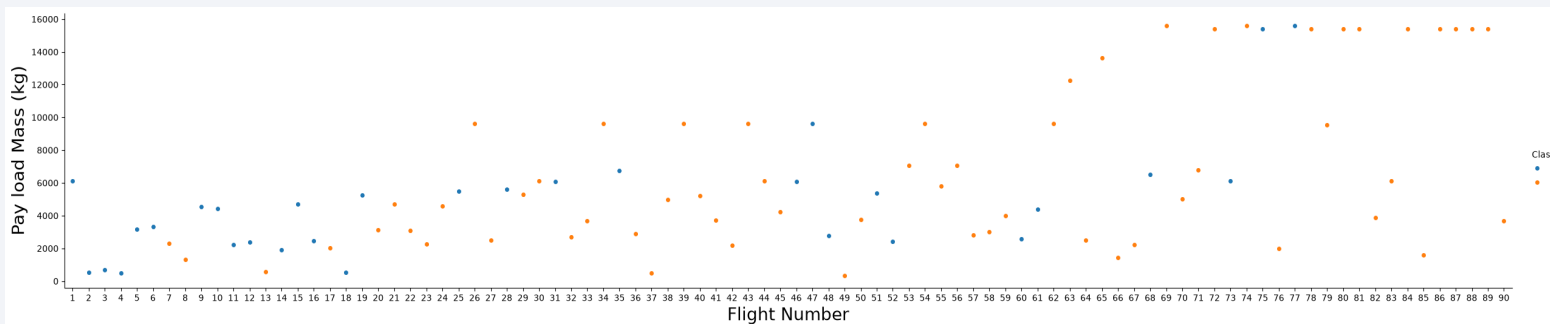
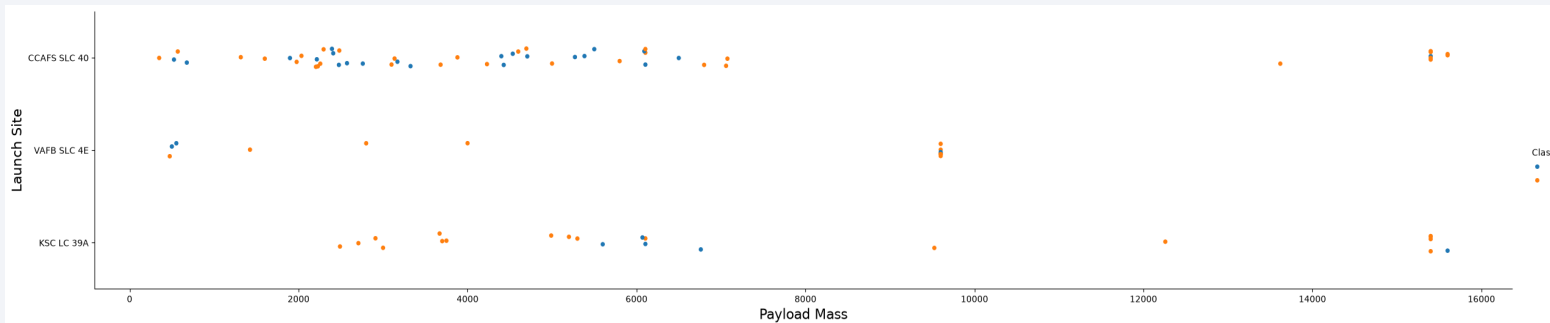
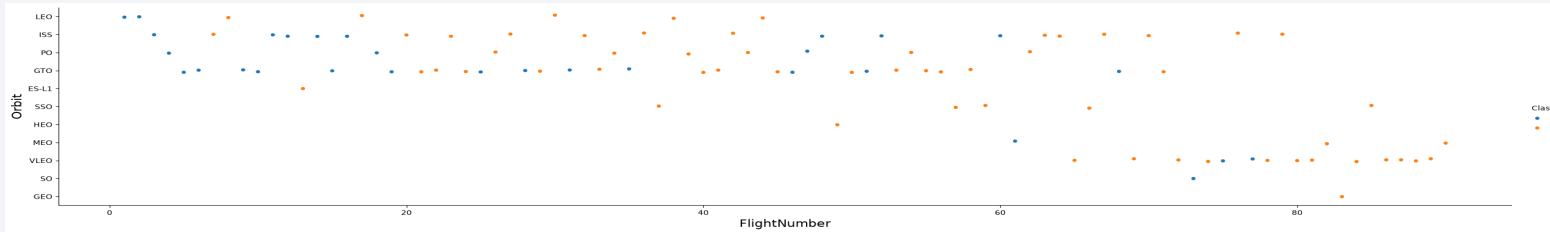
```
df.to_csv('spacex_web_scraped.csv', index=False)
```

Data Wrangling



- https://github.com/HE11032006/project_capstone_data_science_IBM/blob/main/3_Data_Wrangling.ipynb

EDA with Data Visualization



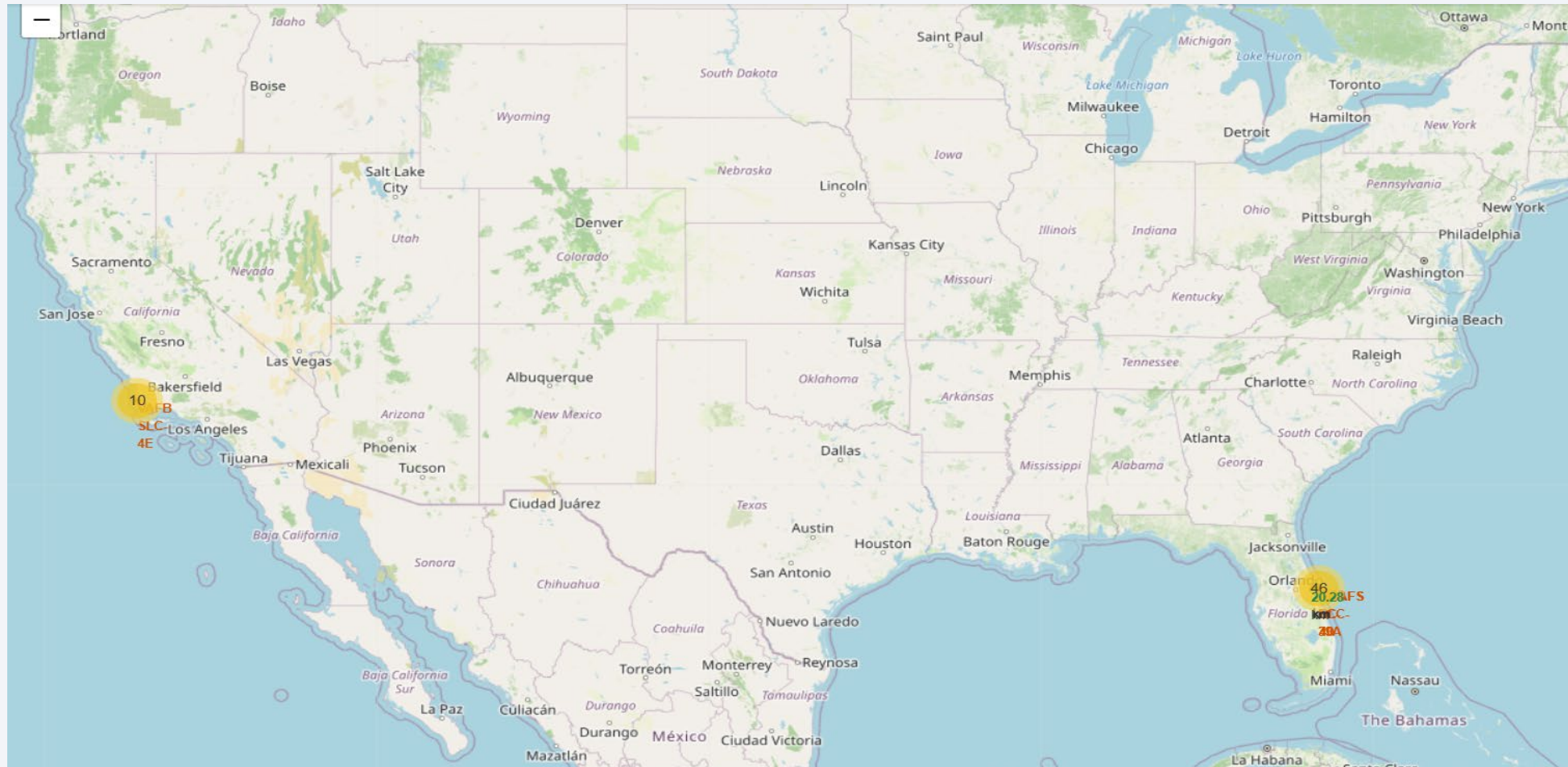
- https://github.com/HE11032006/project_capstone_data_science_IBM/blob/main/4_EDA_Visualization.ipynb

EDA with SQL

- SQL queries

- Displaying the names of the unique launch sites in the space mission
- Displaying 5 records where launch sites begin with the string 'KSC'
- Displaying the total payload mass carried by boosters launched by NASA (CRS)
- Displaying average payload mass carried by booster version F9 v1.1
- Listing the date where the successful landing outcome in drone ship was achieved.
- Listing the names of the boosters which have success in ground pad and have payload mass greater than 4000 but less than 6000
- Listing the total number of successful and failure mission outcomes
- Listing the names of the booster_versions which have carried the maximum payload mass.
- Listing the records which will display the month names, successful landing_outcomes in ground pad ,booster
- List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.
- Ranking the count of successful landing_outcomes between the date 2010 06 04 and 2017 03 20 in descendingorder.
- https://github.com/HE11032006/project_capstone_data_science_IBM/blob/main/5_EDA_Sql.ipynb

Build an Interactive Map with Folium

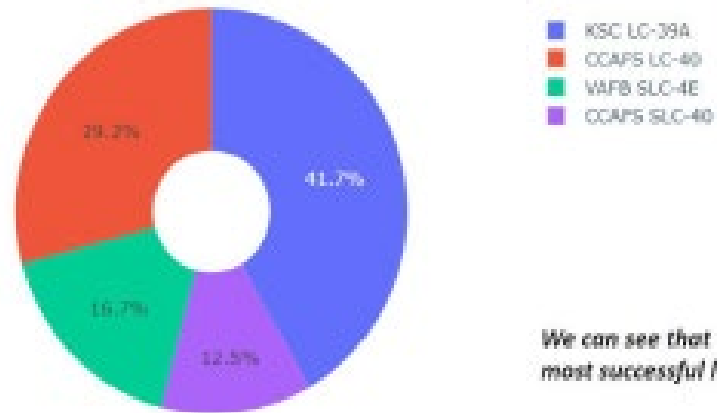


Map markers have been added to the map with aim to finding an optimal location for building a launch site

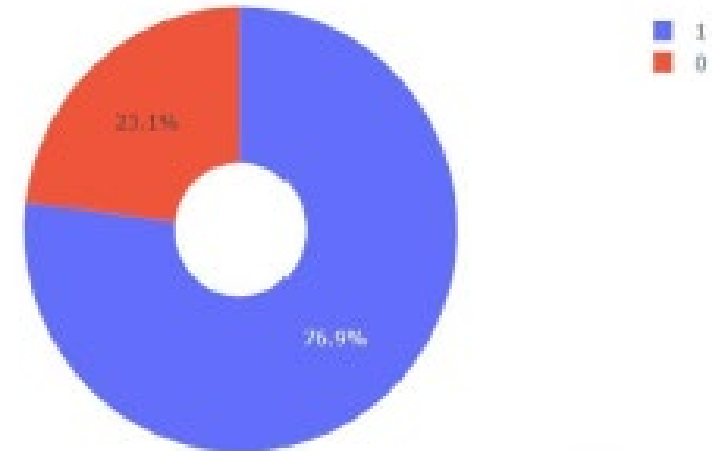
- https://github.com/HE11032006/project_capstone_data_science IBM/blob/main/6_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

Total Success Launches By all sites

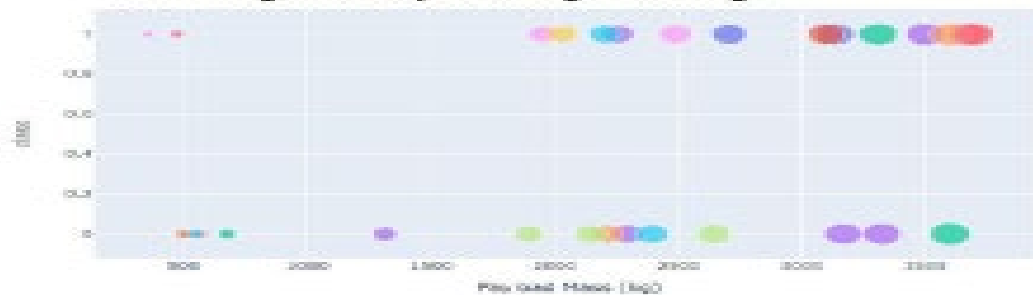


We can see that KSC LC-39A had the most successful launches from all the sites

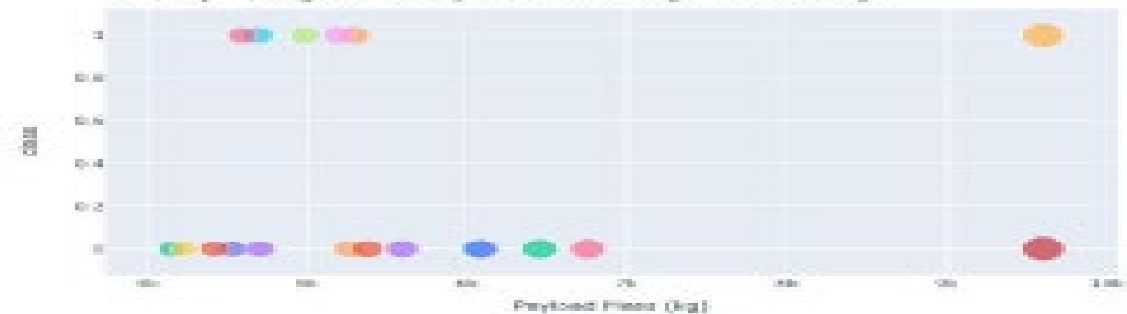


KSC LC-39A achieved a 76.9% success rate while getting a 23.1% failure rate

Low Weighted Payload 0kg - 4000kg



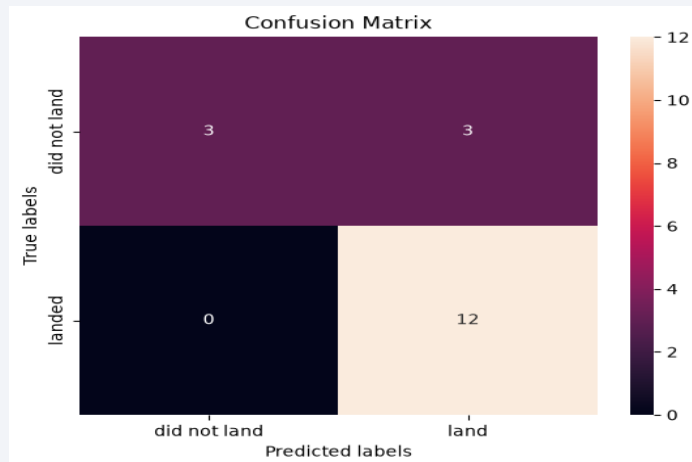
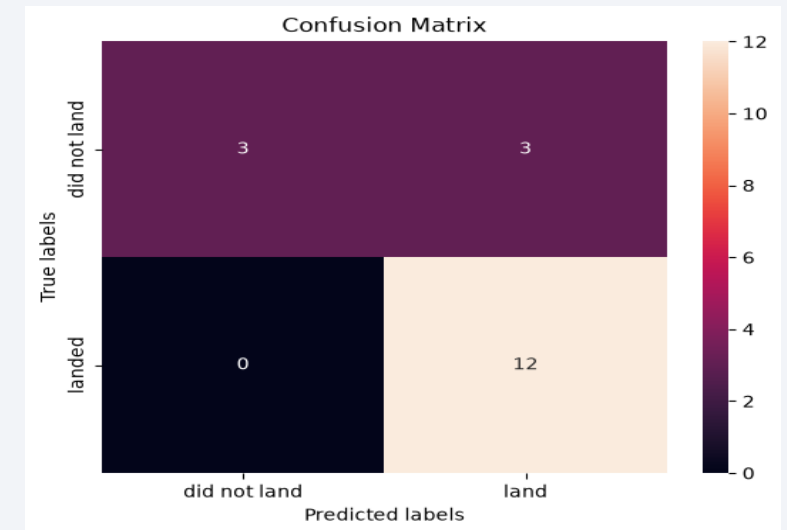
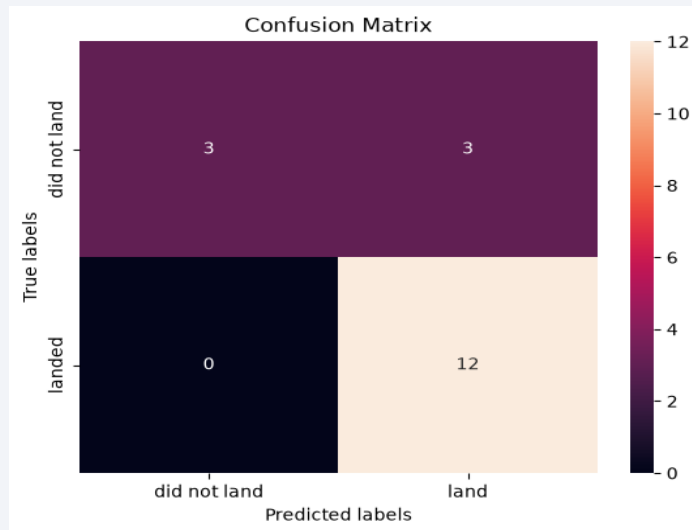
Heavy Weighted Payload 4000kg - 10000kg



We can see the success rates for low weighted payloads is higher than the heavy weighted payloads

Predictive Analysis (Classification)

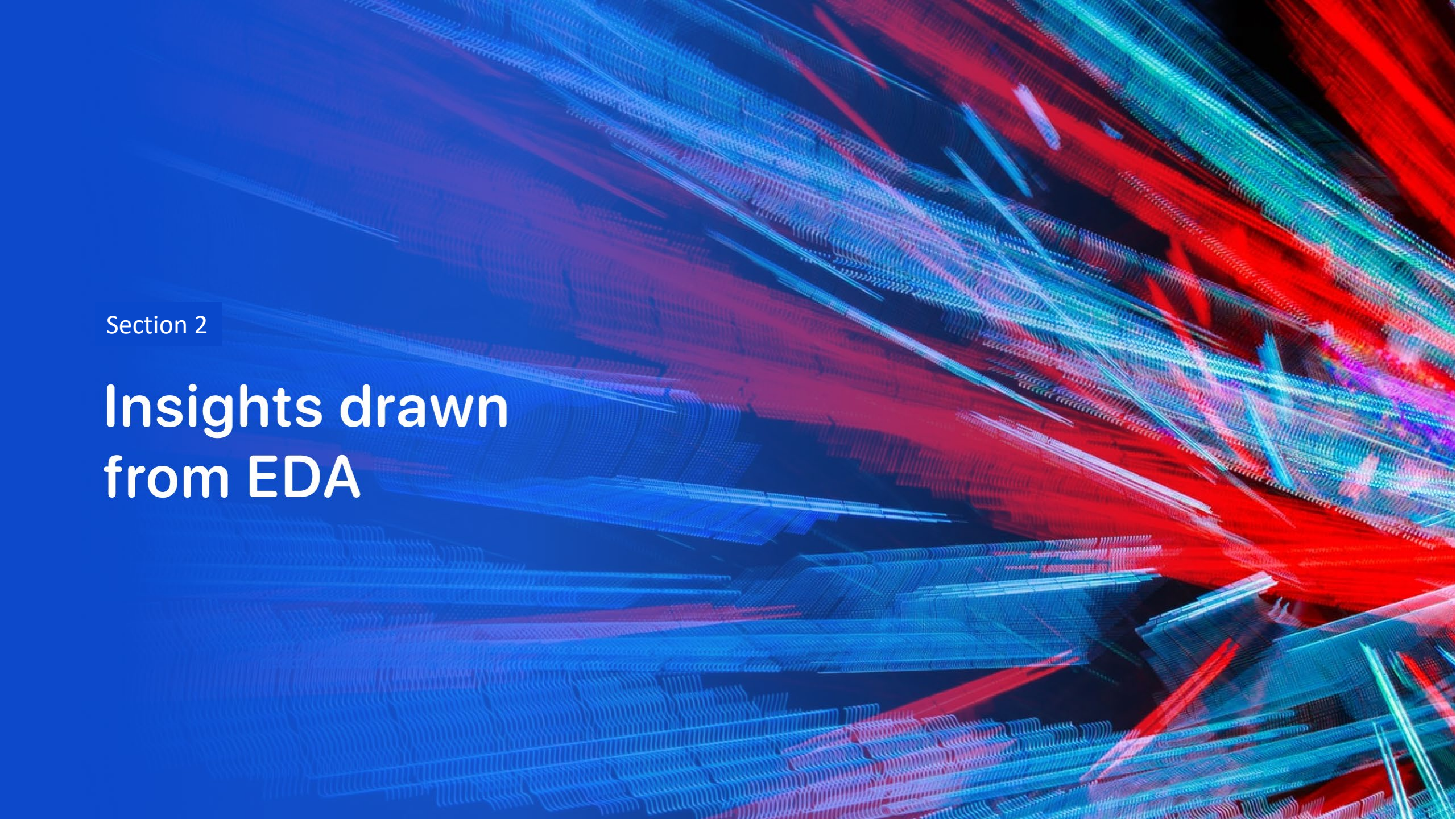
- The SVM, KNN, and Logistic Regression model achieved the highest accuracy at 83.3%.



https://github.com/HE11032006/project_capstone_data_science_IBM/blob/main/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

Results

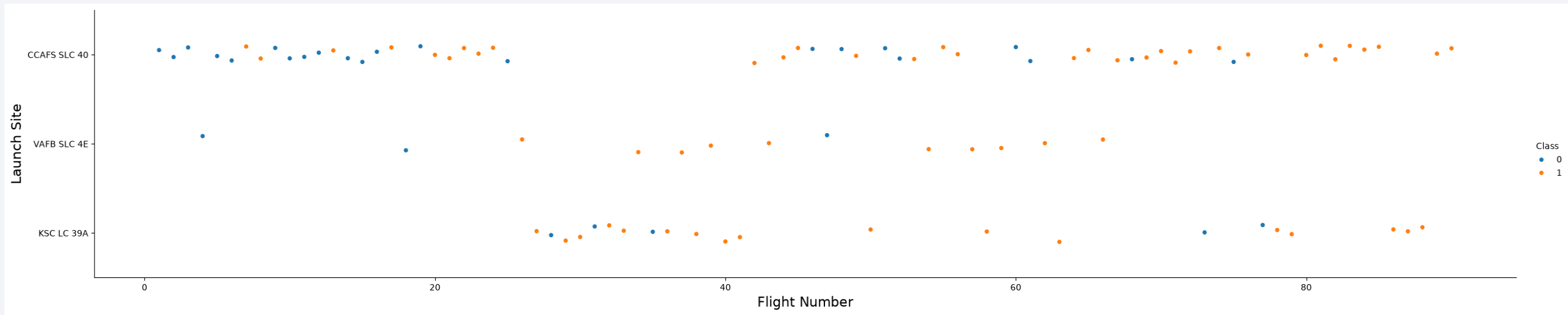
- The SVM, KNN, and Logistic Regression models are the best in terms of prediction accuracy for this dataset.
- Low weighted payloads perform better than the heavier payloads.
- The success rates for SpaceX launches is directly proportional time in years they will eventually perfect the launches.
- KSC LC 39A had the most successful launches from all the sites.
- Orbit GEO,HEO,SSO,ES L1 has the best Success Rate.

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

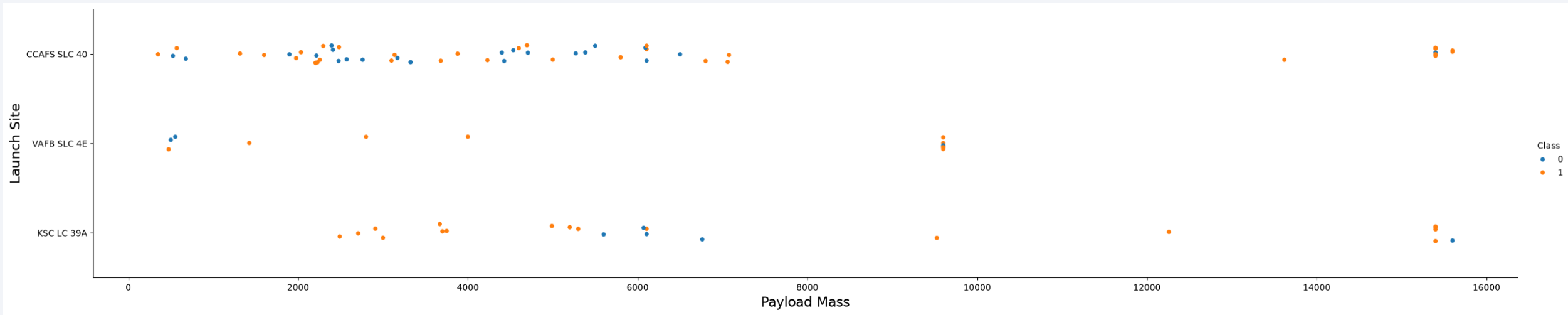
Insights drawn from EDA

Flight Number vs. Launch Site



- Flight Number vs. Launch Site

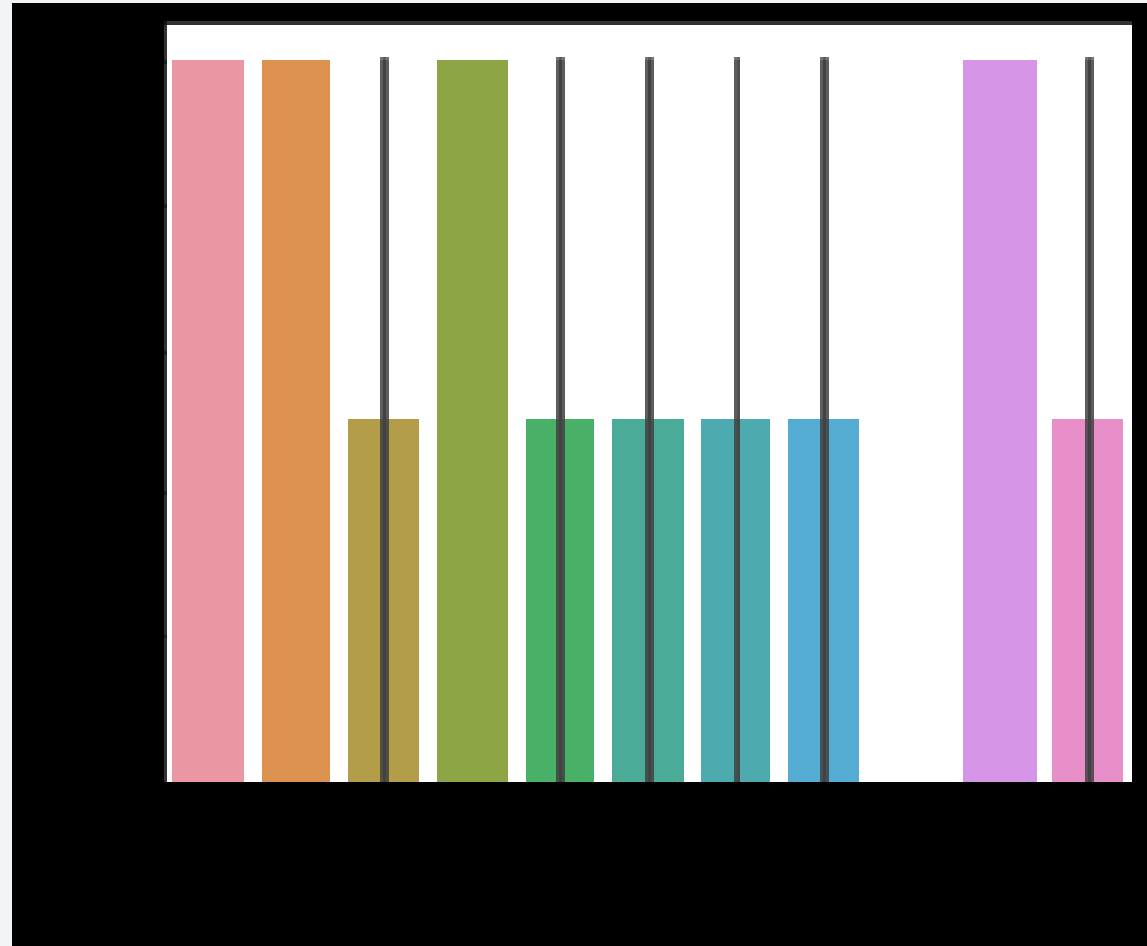
Payload vs. Launch Site



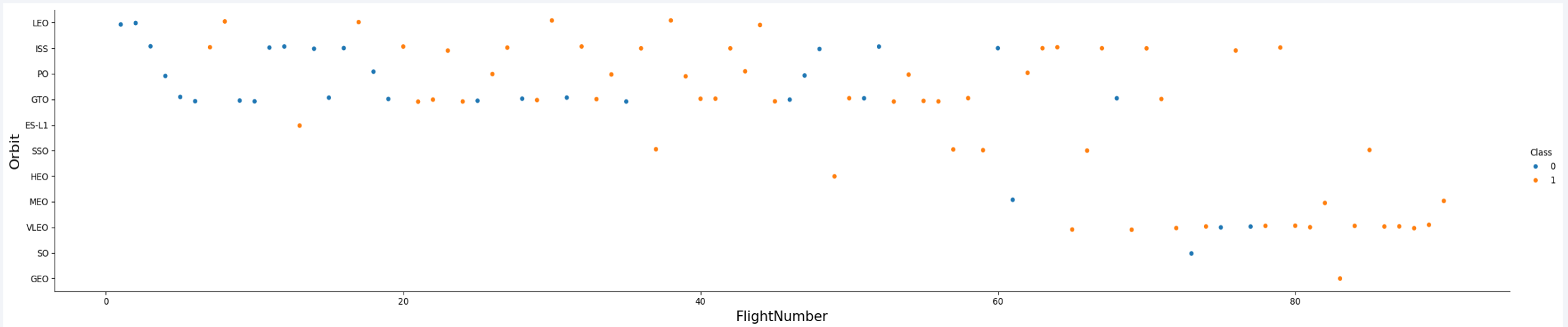
- Payload vs. Launch Site

Success Rate vs. Orbit Type

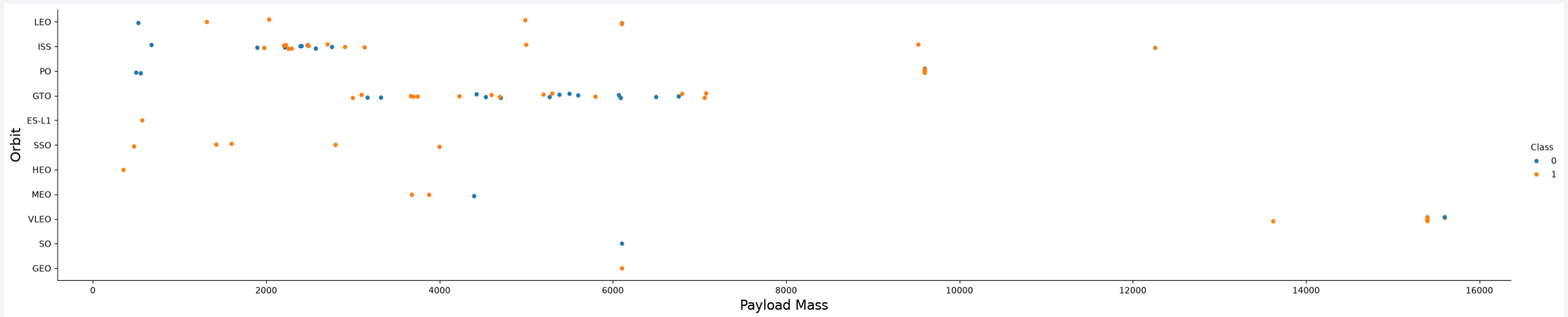
- Bar chart for the success rate of each orbit type



Flight Number vs. Orbit Type

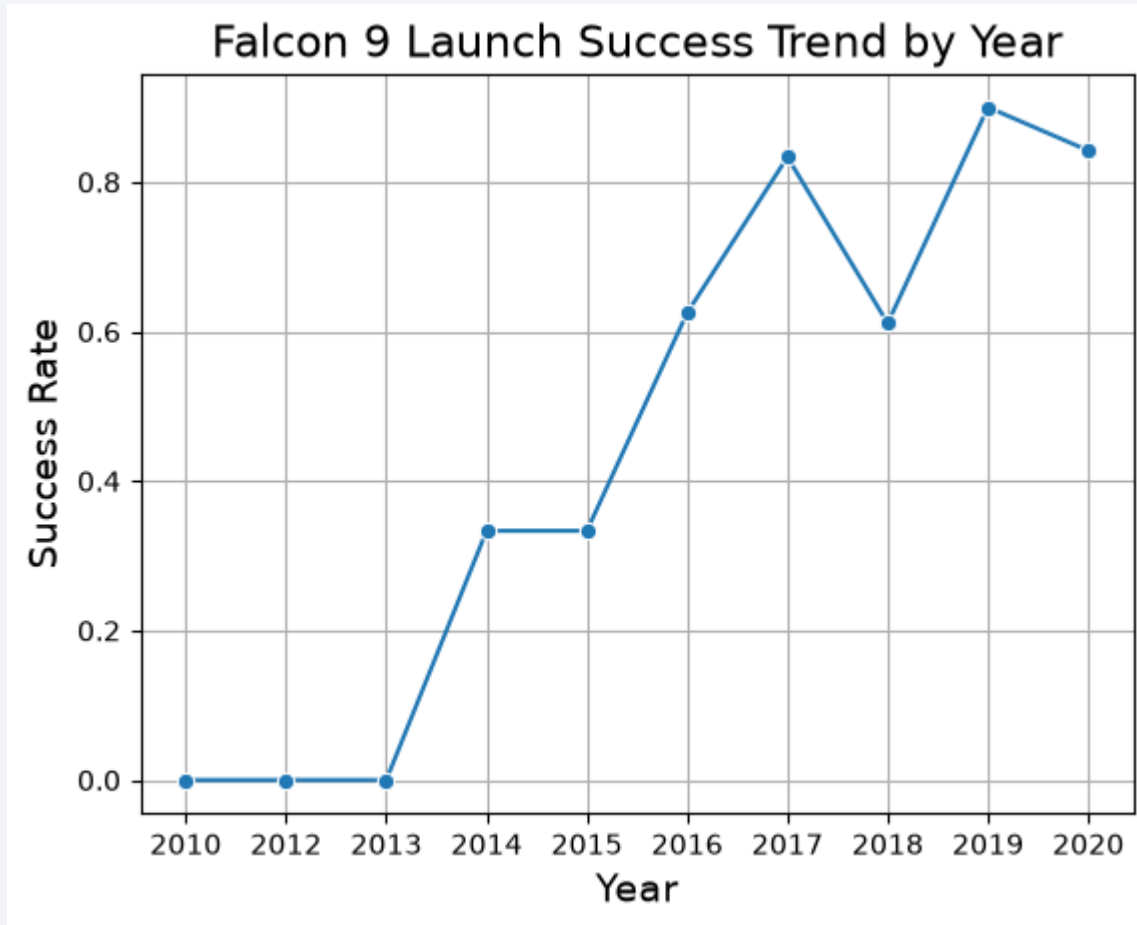


Payload vs. Orbit Type



- Scatter point of payload vs. orbit type

Launch Success Yearly Trend



- Line chart of yearly average success rate

All Launch Site Names

- %sql select distinct(LAUNCH_SITE) from SPACEXTBL

launch_site

CCAFS LC-40

CCAFS SLC-40

KSC LC-39A

VAFB SLC-4E

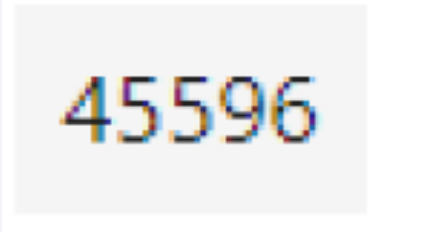
Launch Site Names Begin with 'CCA'

- %sql select * from SPACEXTBL where LAUNCH_SITE like 'CCA%' limit 5

DATE	time_utc	booster_version	launch_site	payload	payload_mass_kg	orbit	customer	mission_outcome	landing_outcome
2010-05-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- %sql select sum(PAYLOAD_MASSKG_) from SPACEXTBL where CUSTOMER = 'NASA (CRS)'



45596

Average Payload Mass by F9 v1.1

- %sql select avg(PAYLOAD_MASSKG_) from SPACEXTBL where BOOSTER_VERSION = 'F9 v1.1'



2928.400000

First Successful Ground Landing Date

- %sql select min(DATE) from SPACEXTBL where Landing Outcome = 'Success (ground pad)'

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- %sql select BOOSTER_VERSION from SPACEXTBL where Landing Outcome = 'Success (drone ship)' and PAYLOAD_MASS KG_ > 4000 and PAYLOAD_MASS KG_ < 6000

booster_version

F9 FT B1022

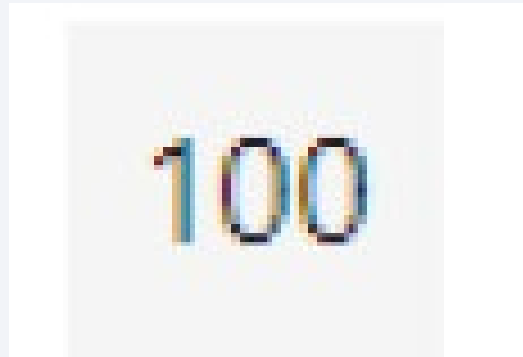
F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- %sql select count(MISSION_OUTCOME) from SPACEXTBL
where MISSION_OUTCOME = 'Success' or MISSION_OUTCOME = 'Failure (in flight)'



100

Boosters Carried Maximum Payload

- %sql select BOOSTER_VERSION from SPACEXTBL where PAYLOAD_MASS KG_ = (select max(PAYLOAD_MASSKG_) from SPACEXTBL)

```
booster_version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7
```

2015 Launch Records

- %sql select * from SPACEXTBL where Landing Outcome like 'Success%' and (DATE between '2015-01-01' and '2015-12-31') order by date desc

time_utc	booster_version	launch_site	payload	payload_mass_kg	orbit	customer	mission_outcome	landing_outcome
14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
17:54:00	F9 FT B1029.1	VAFB SLC-4E	Iridium NEXT 1	9600	Polar LEO	Iridium Communications	Success	Success (drone ship)
05:26:00	F9 FT B1026	CCAFS LC-40	JCSAT-16	4600	GTO	SKY Perfect JSAT Group	Success	Success (drone ship)
04:45:00	F9 FT B1025.1	CCAFS LC-40	SpaceX CRS-9	2257	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
21:39:00	F9 FT B1023.1	CCAFS LC-40	Thaicom 8	3100	GTO	Thaicom	Success	Success (drone ship)
00:00:00	F9 FT B1022.1	CCAFS LC-40	SKY Perfect JSAT	4600	GTO	SKY Perfect JSAT	Success	Success (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- %sql select * from SPACEXTBL where Landing Outcome like 'Success%' and (DATE between '2010-06-04' and '2017-03-20') order by date desc

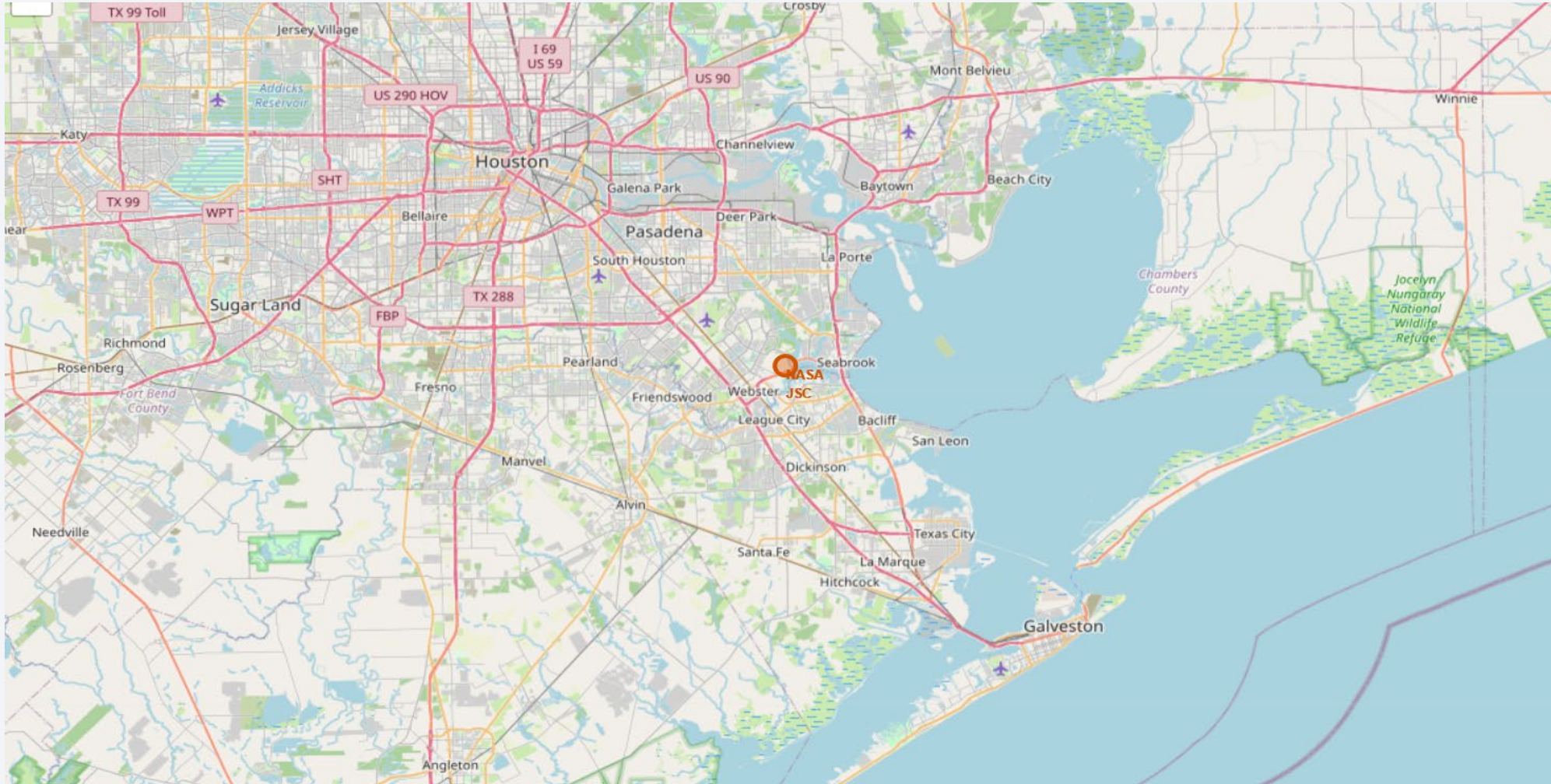
2016-05-27	21:39:00	F9 FT B1023.1	CCAFS LC-40	Thaicom 8	3100	GTO	Thaicom	Success	Success (drone ship)
2016-05-06	05:21:00	F9 FT B1022	CCAFS LC-40	JCSAT-14	4696	GTO	SKY Perfect JSAT Group	Success	Success (drone ship)
2016-04-08	20:43:00	F9 FT B1021.1	CCAFS LC-40	SpaceX CRS-8	3136	LEO (ISS)	NASA (CRS)	Success	Success (drone ship)
2015-12-22	01:29:00	F9 FT B1019	CCAFS LC-40	OG2 Mission 2 11 Orbcomm-OG2 satellites	2034	LEO	Orbcomm	Success	Success (ground pad)

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a curved line separating the dark surface from the deep blue of space.

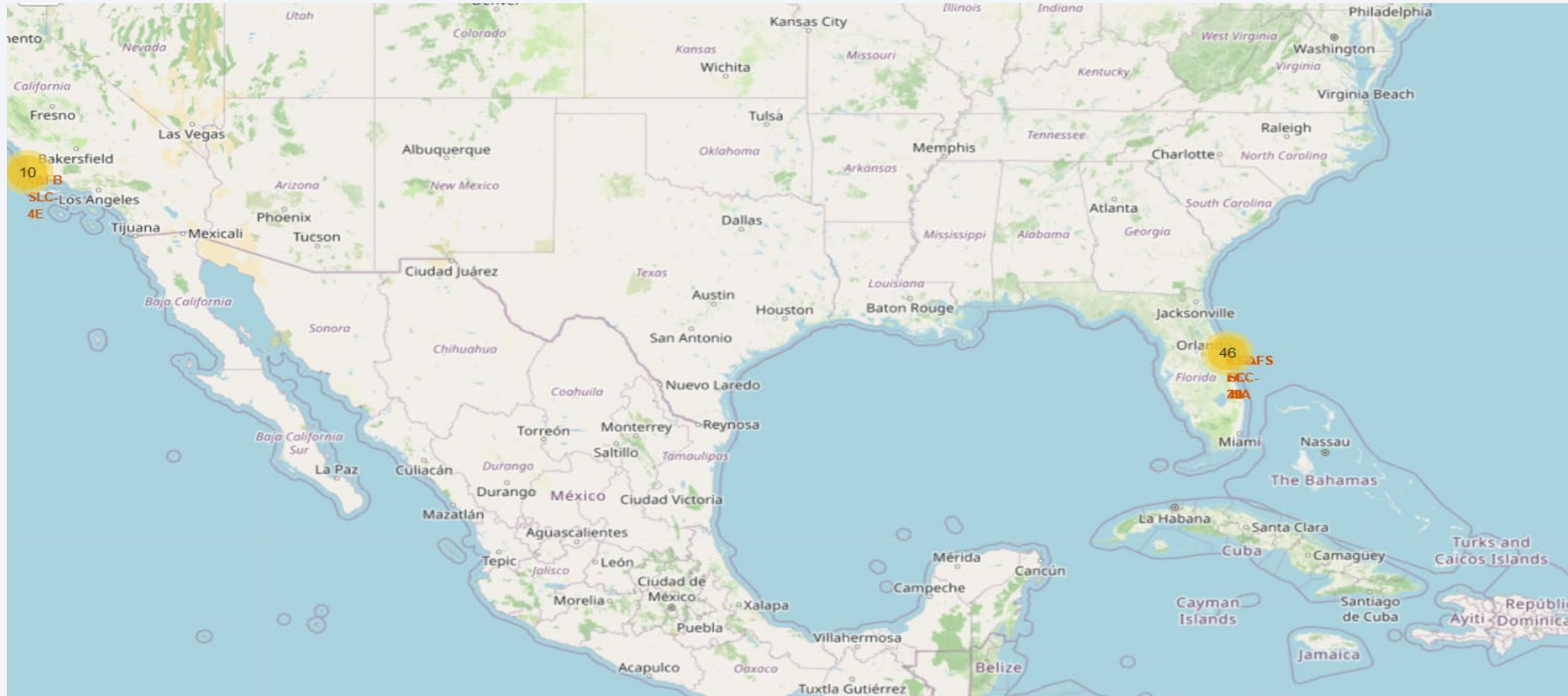
Section 3

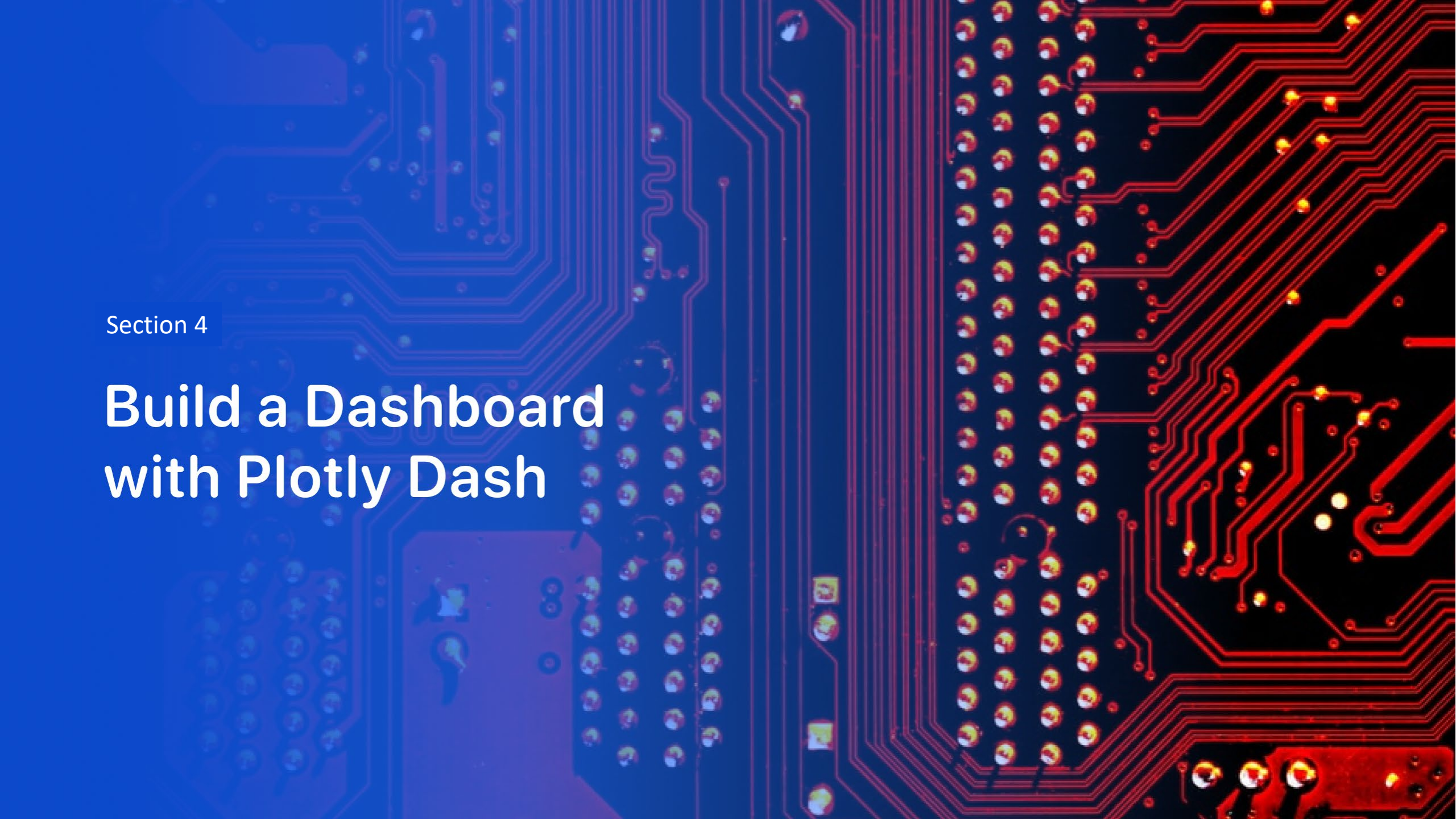
Launch Sites Proximities Analysis

Each sites



Success/failed launches marked on the map



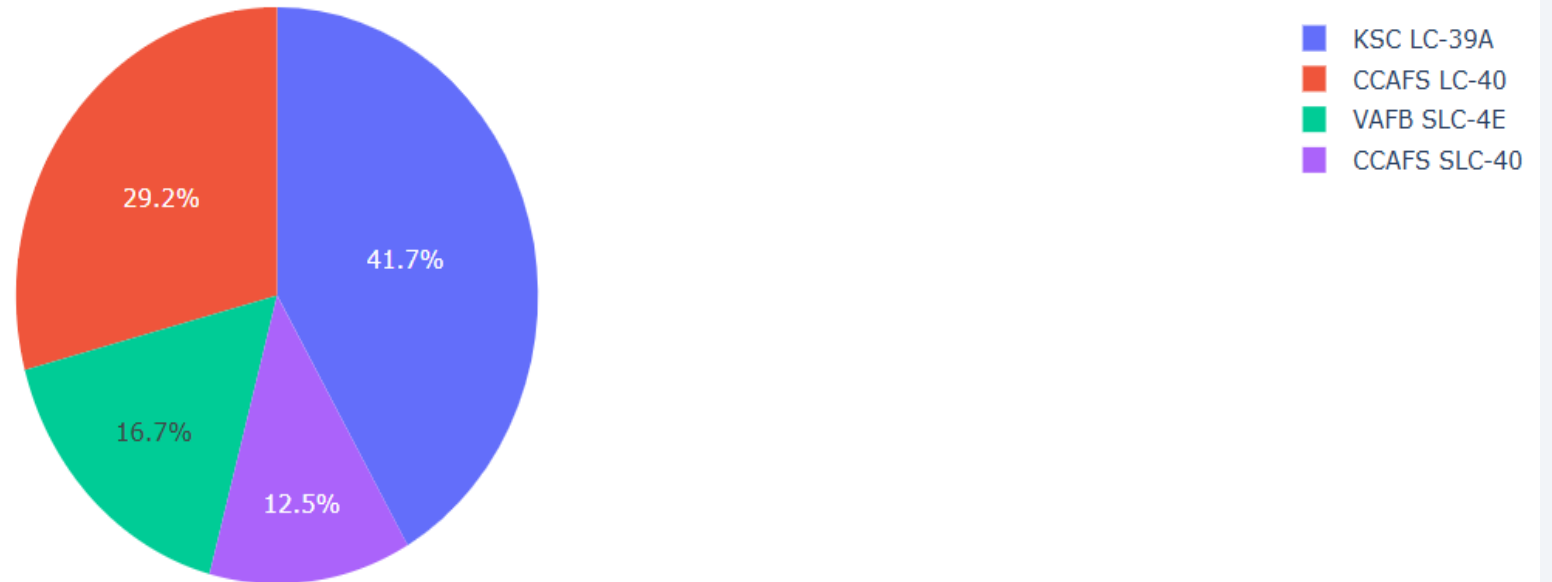


Section 4

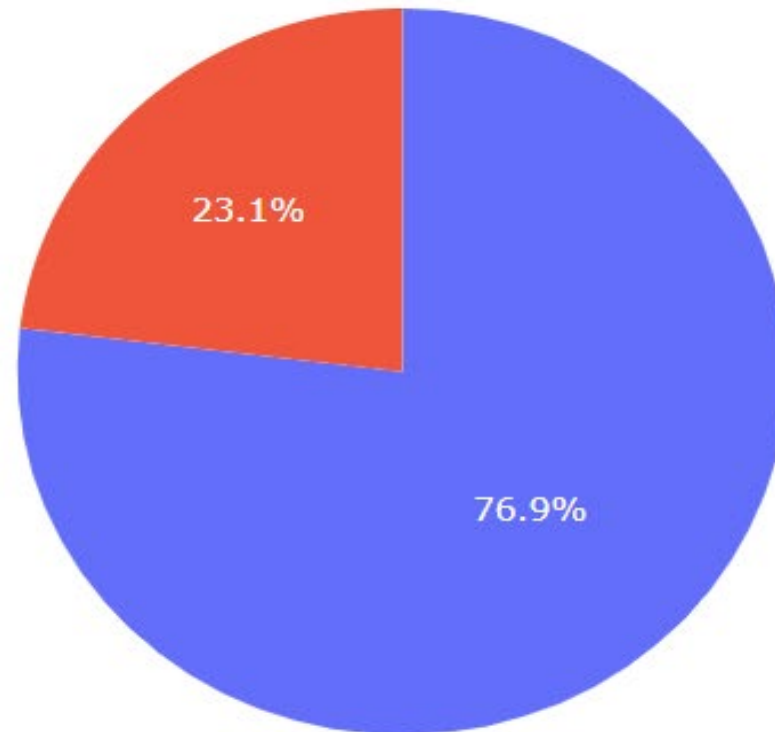
Build a Dashboard with Plotly Dash

Total success launches by all sites

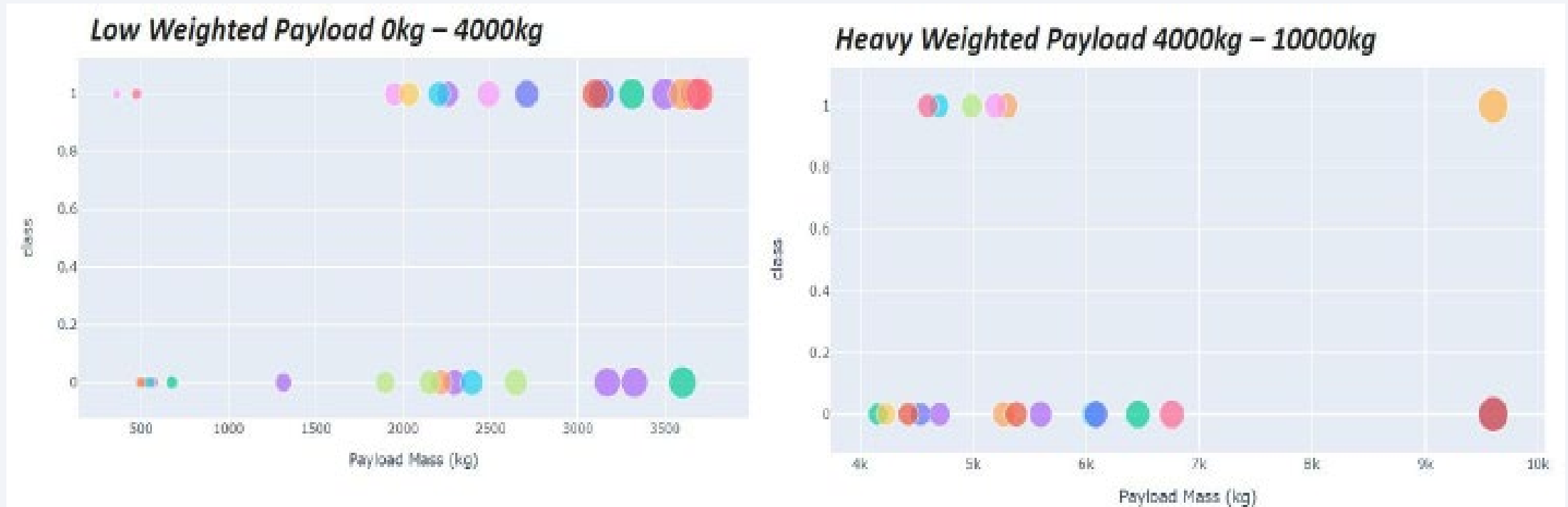
Total Successful Launches by Site



Success rate by site



Payload vs launch outcome

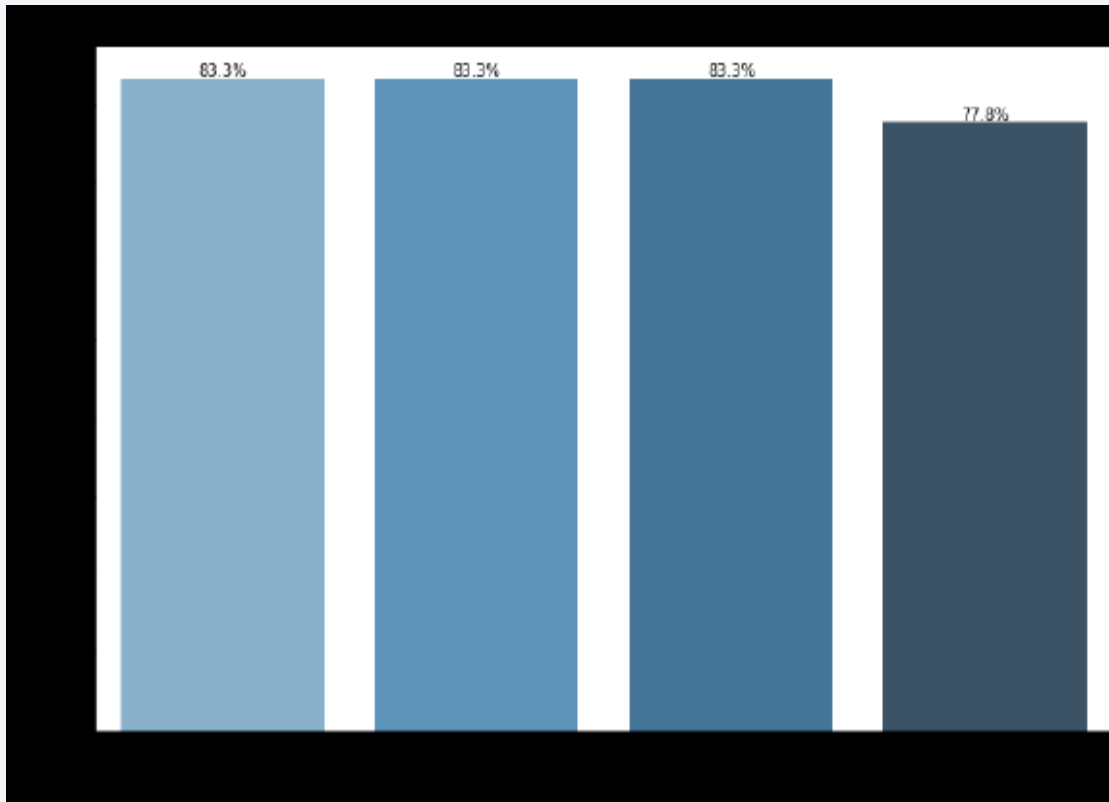


We can see the success rates for low weighted payloads is higher than the heavy weighted payloads

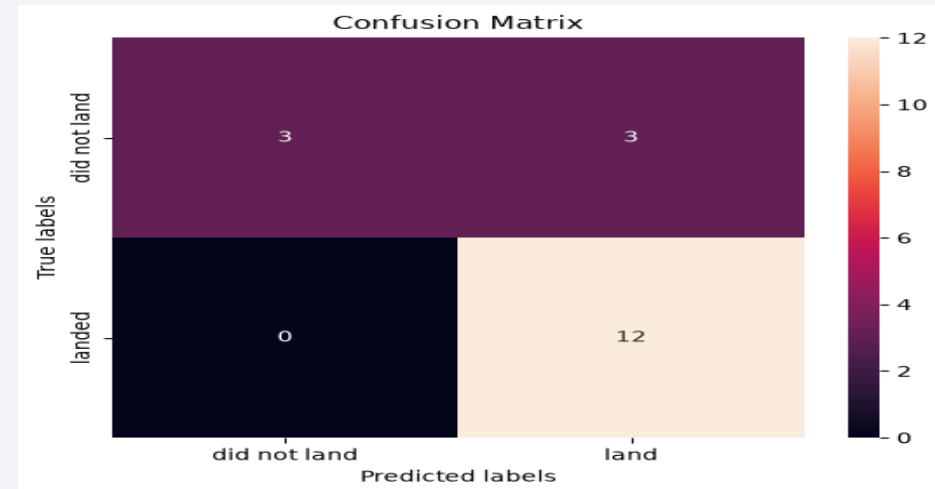
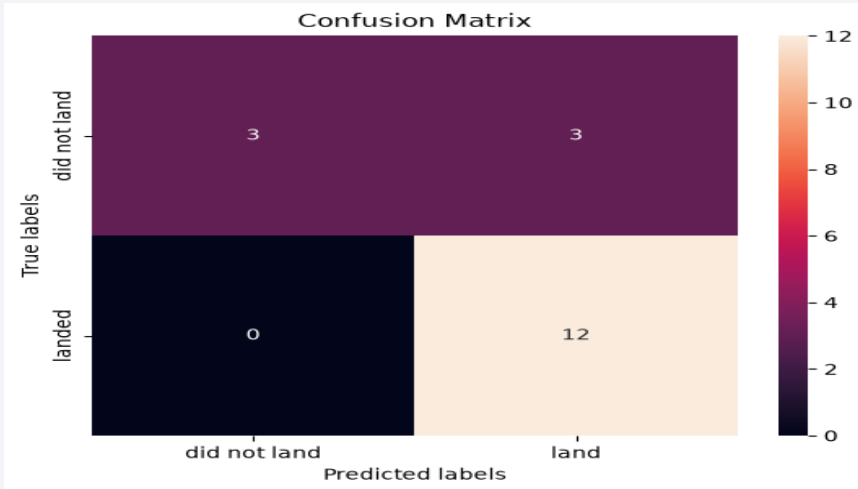
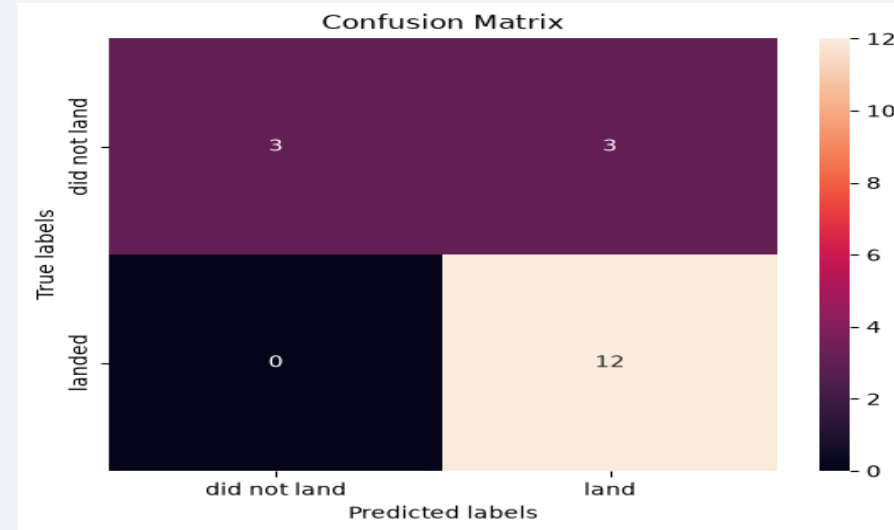
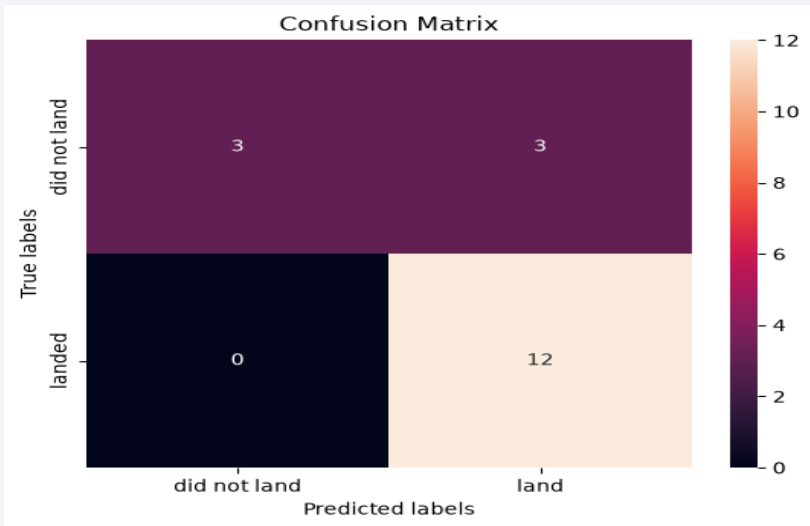
Section 5

Predictive Analysis (Classification)

Classification Accuracy



Confusion Matrix



Conclusions

- The SVM, KNN, and Logistic Regression models are the best in terms of prediction accuracy for this dataset.
- Low weighted payloads perform better than the heavier payloads.
- The success rates for SpaceX launches is directly proportional time in years they will eventually perfect the launches.
- KSC LC 39A had the most successful launches from all the sites.
- Orbit GEO,HEO,SSO,ES L1 has the best Success Rate.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

